

Beyond Recall: Behavioral Specification as the Missing Primitive for AI Personalization

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github.com/agulaya24/base-layer · github.com/agulaya24/memory-study-repo

Disclosure: The author is the founder of Base Layer, which provides an open-source implementation of the pipeline described in this paper. All data, code, and evaluation artifacts are released under Apache 2.0.

Abstract

In order to have aligned AI memory and actions—where alignment means the AI’s behavior accords with how a specific person reasons and decides—AI requires a representation of how its user thinks. Current memory systems store what someone said. Preference models store what someone likes. Neither captures how someone reasons.

We introduce **Behavioral Specifications**: a compressed document ($\sim 5K$ tokens) encoding how a person thinks, decides, and reasons. It is generated from source text by extracting behavioral facts using 47 constrained predicates, authoring three interpretive layers (anchors, core, predictions), and composing them into a unified specification.

We tested this across 14 subjects from 11 cultures, 6 response models, 4 state-of-the-art commercial memory systems (Mem0, Letta, Supermemory, Zep), 3 providers, and 7 calibrated LLM judges. The behavioral specification significantly improves representational accuracy for subjects with low pretraining representation (12 of 14 subjects, Wilcoxon signed-rank $p = 0.0015$), with improvements ranging from +6% to +136% over baseline.

Key findings:

1. Adding a behavioral specification to 4 SOTA memory systems increases representational accuracy by +12% to +66% (Letta +45%, Mem0 +22%, Supermemory +12%, Zep +66%).
2. A 5,250-word ($\sim 8.6K$ token) spec outperforms 25,000 words ($\sim 33K$ tokens) of raw source text by 29% normalized.
3. For low-representation subjects (baseline below 2.4/5.0), improvement ranges from +6% to +136%. For well-represented subjects, the specification is unnecessary or slightly harmful.
4. Four SOTA systems have zero top-1 retrieval overlap 68% of the time given the same facts and the same question. The specification still improves every system.
5. A wrong behavioral specification scores near baseline across all 14 subjects. The correct content drives the improvement, not the presence of a framework.

Seven judges confirm the same condition rankings (pairwise Spearman $\rho = 0.89\text{--}0.98$). Krippendorff’s $\alpha = 0.723$.

1 Introduction

In order to have aligned AI memory and actions—where alignment means the AI’s behavior accords with how a specific person reasons and decides—AI requires a representation of how its

user thinks. Current memory systems store what someone said. Preference models store what someone likes. Neither captures how someone reasons.

We call this a **Behavioral Specification**: a compressed document (~5K tokens) that encodes a person’s decision patterns, values under conflict, risk tolerance, learning style, and reasoning structure. It is not a preference model, not a fact store, and not a persona. It is a model of how someone reasons about their experience, generated from source text through a fully automated pipeline (Section 3.3).

The definitive memory benchmark, LongMemEval (?) (ICLR 2025), tests recall: “What did the user say about X in conversation 47?” Four state-of-the-art commercial memory systems (Mem0, Letta, Supermemory, Zep) all score 85%+ on these retrieval tasks. But recall and preference storage share the same limitation: they capture what was expressed, not the reasoning behind it.

To illustrate the gap: when we gave four SOTA memory systems the same 462 facts and the same question, they retrieved completely different top-1 facts 68% of the time (39% at top-3, 22% at top-5). They all pass recall benchmarks, but cannot agree on what is relevant.

1.1 1.1 What We Mean by Alignment

In this study, “alignment” refers to *behavioral alignment*: making an AI’s actions accord with a specific person’s reasoning patterns, values, and decision-making. We ask: can the AI act the way you would act, given how you think?

1.2 1.2 What the Specification Captures

The Behavioral Specification identifies *interpretive patterns*: durable structures of reasoning that show up over time. A memory system that stores “his father was violent (biographical, from Hamerton’s autobiography)” has captured a fact. A specification that captures “this person evaluates authority figures on two simultaneous ledgers, virtue and failure, refusing to collapse them into a single verdict (behavioral, derived from the same source)” has captured the interpretive pattern that gives that fact its significance. The pattern was never stated. It was derived from dozens of facts. This is what makes it a specification rather than a snapshot.

2 4. Results (Excerpt)

2.1 4.4 The Global Gradient: N=14 Across 11 Cultures

3 8. Conclusion

The problem is not recall. Four memory systems score 85%+ on the same benchmark. Given the same facts and the same question, they retrieve completely different facts 68% of the time. They have solved storage. They have not solved understanding.

The problem is interpretation. A fact does not carry its own significance. The lens through which it is evaluated is what gives it personal significance. No memory system captures that lens. The behavioral specification does.

The specification provides for private individuals what pretraining provides for public figures. It does so with full traceability. This is what separates personalization from profiling.

Table 1: Global gradient: specification impact across 14 subjects. Scores are means across 6 judges. Effect: percentage improvement of Spec over Baseline.

Subject	Culture	Baseline	Spec	Wrong	Facts	F+Spec	Effect	Abs.
Hamerton	British	1.25	2.94	1.79	2.55	3.08	+136%	+1.69
Sunity Deveen	Indian	1.03	2.27	1.29	2.46	2.41	+121%	+1.24
Georg Ebers	German	1.04	1.80	1.50	2.21	2.34	+73%	+0.76
Fukuzawa	Japanese	1.80	2.56	2.11	2.89	2.99	+42%	+0.75
Mary Seacole	Caribbean	1.77	2.48	1.43	2.63	2.60	+40%	+0.71
Bernal Diaz	Latin American	1.83	2.53	2.13	2.71	2.78	+38%	+0.70
Keckley	Black American	1.92	2.64	1.50	2.57	2.62	+38%	+0.73
Yung Wing	Chinese	1.88	2.22	2.20	2.13	2.40	+18%	+0.34
Rousseau	French	2.44	2.81	1.91	2.32	2.53	+15%	+0.37
Babur	Central Asian	2.03	2.28	1.23	2.37	2.39	+12%	+0.25
Cellini	Italian	2.56	2.72	1.94	2.61	2.80	+6%	+0.16
Augustine	North African	2.80	2.97	2.54	3.09	3.22	+6%	+0.17
Equiano	West African	2.93	2.70	2.18	2.63	2.65	-8%	-0.24
Zitkala-Sa	Native American	2.34	2.03	1.66	2.10	2.02	-13%	-0.31

Wilcoxon signed-rank: $p = 0.0015$, $N = 14$. Krippendorff's $\alpha = 0.723$. Effect: relative improvement over baseline raw score. Abs.: absolute score difference.

Memory systems store what was said. Preference models store what was liked. The behavioral specification captures how someone makes sense of what happened to them. That is the primitive that has been missing.

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